

Fire Fighter Robot with Obstacle Avoidance

Course Title : Robotics Lab

Course Code : CSE 438

Submitted To:

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Presented By



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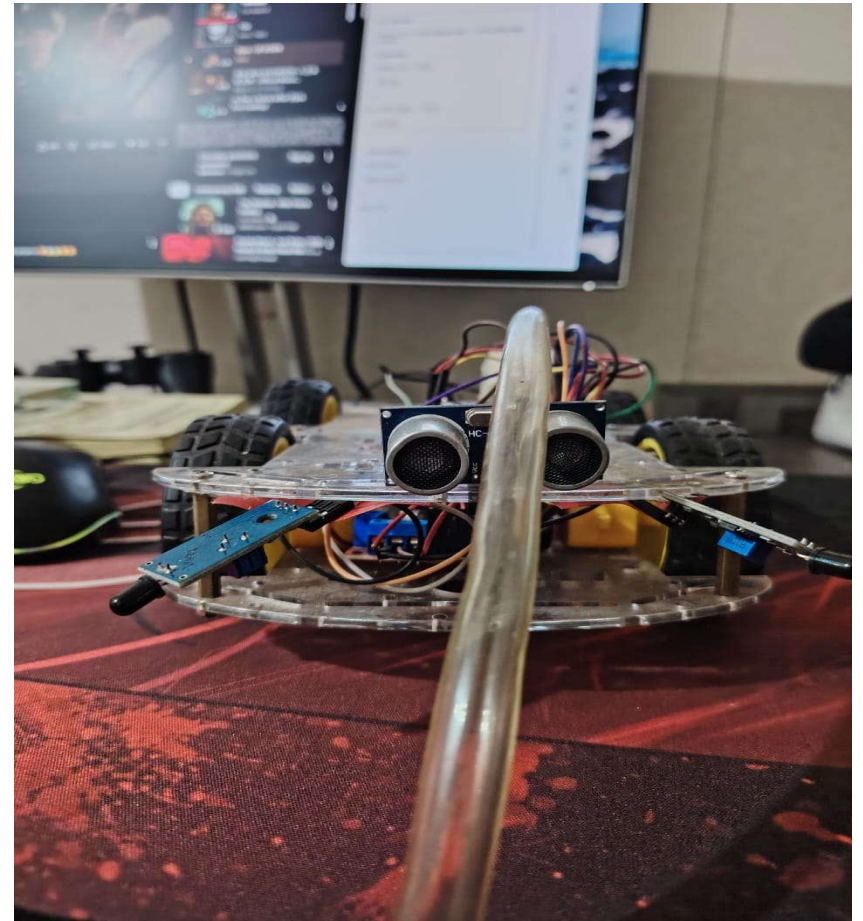
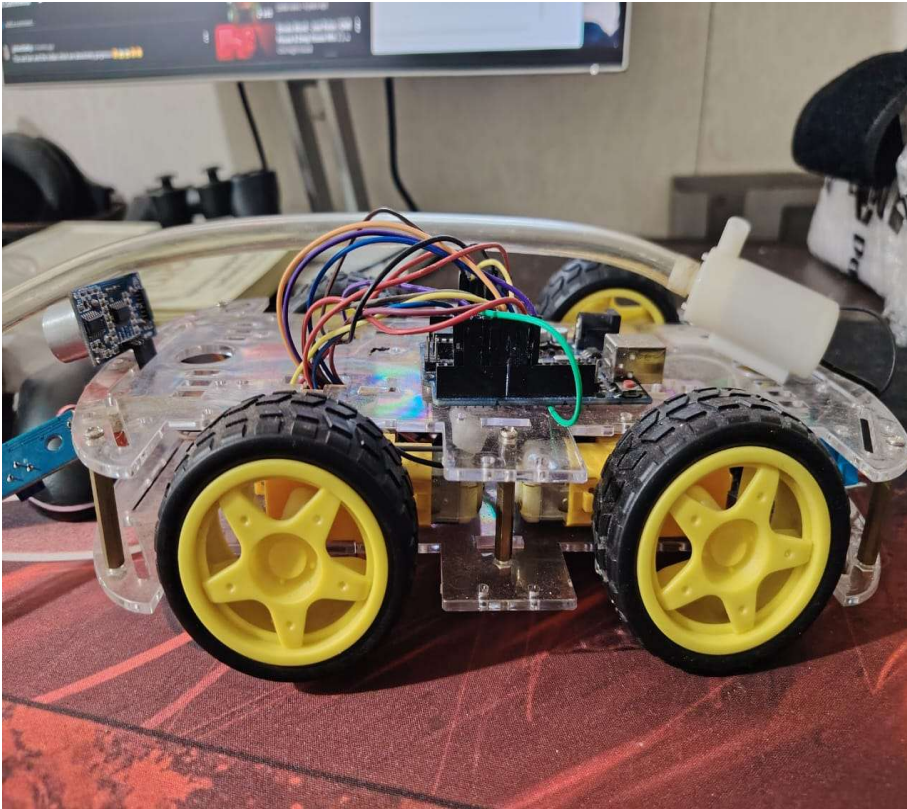
Project Objective

The objective of this project is to design and develop an autonomous fire fighter robot capable of detecting fire, avoiding obstacles, and extinguishing flames automatically. The robot uses sensors and embedded systems to improve safety and reduce human involvement in dangerous fire situations.

Project Goals

- Detect fire automatically using flame sensors
 - Avoid obstacles during movement
 - Extinguish fire efficiently using a pump
 - Reduce human risk in dangerous environments
 - Develop a low-cost and reliable robotic system
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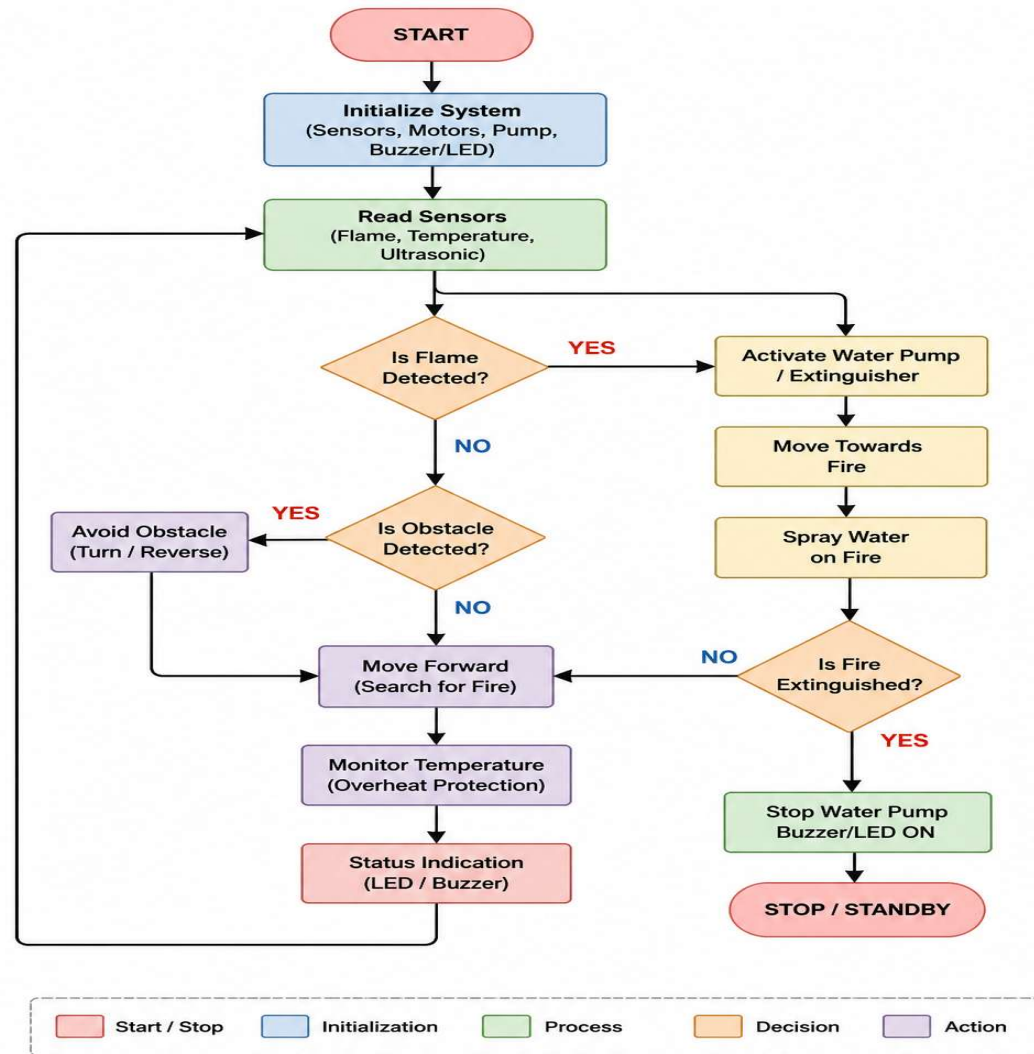
Fire Fighter Robot



EQUIPMENTS

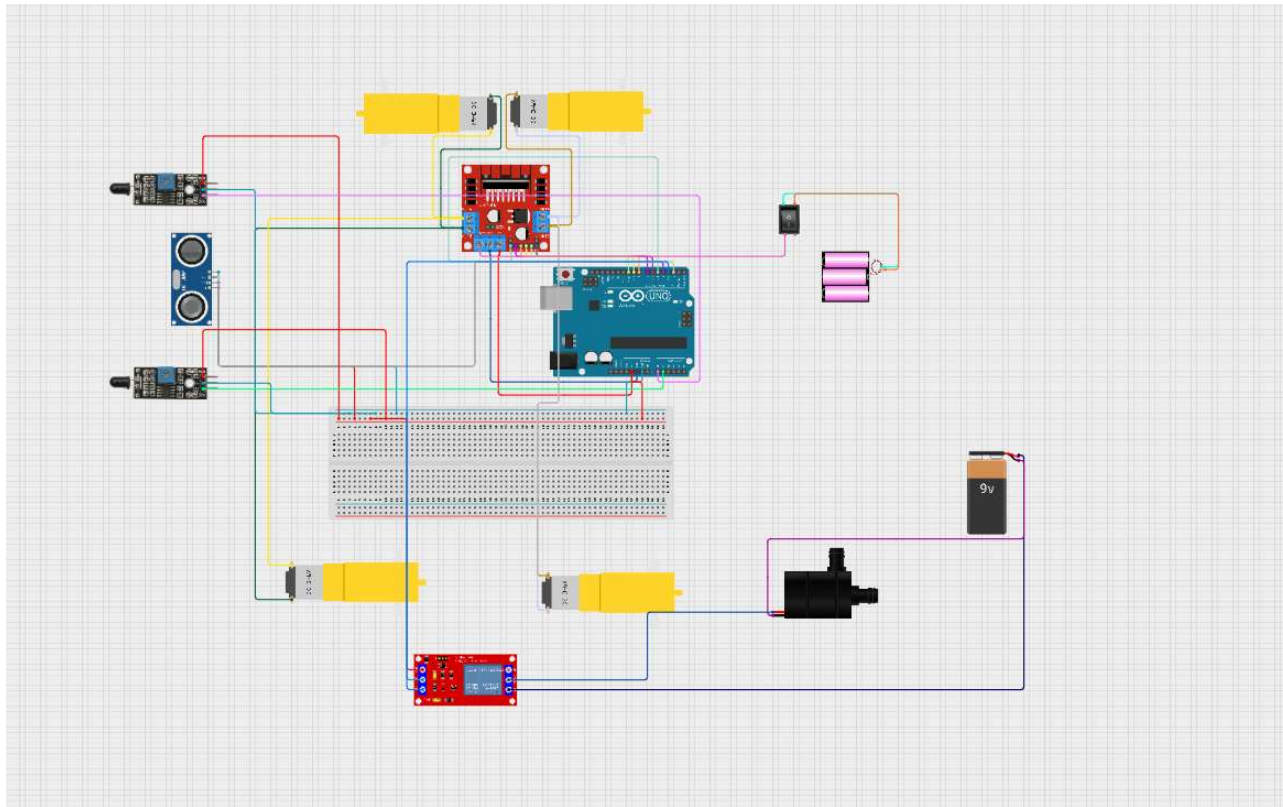
No.	Component Name	Quantity
1	Arduino UNO (clone)	1
2	Motor Driver Module (L298N / L293D)	1
3	DC Geared Motors (x4 for wheels)	4
4	Robot Car Chassis (with wheels)	1
5	Water Pump (mini submersible pump or spray pump)	1
6	Water/Fire Container + Pipe Setup	1
7	Flame Sensor Module	2
8	Ultrasonic Sensor (for obstacle avoidance, optional)	1
9	Battery Pack (Rechargeable Li-ion or 12V lead-acid)	2
10	Breadboard, Switches, Misc. Components	1
11	Jumper Wires	-

Flow Diagram



FIREFIGHTER ROBOT FLOW DIAGRAM

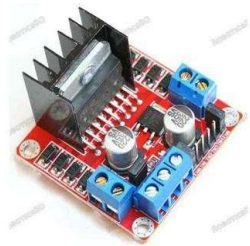
Virtual Diagram



HARDWARE IMAGES



ESP32



Motor Driver



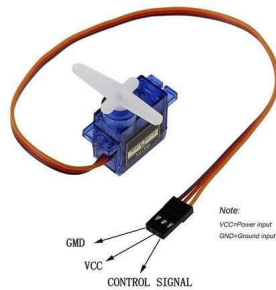
IR Flame Sensor



Ultrasonic Sensor



**Temperature &
Humidity Sensor**



Servo Motor



**Submersible Water
Pump**



**4WD Smart Robot
Chassis Kit**

HARDWARE IMAGES



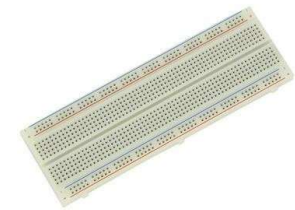
ESP32-CAM Module



Li-Ion Battery



Battery Case



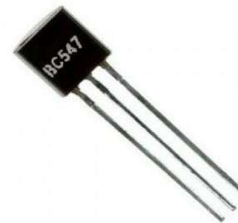
Solder-less Breadboard



5V Single Channel Relay Module



Buck Converter



BC547 Transistor



4WD Smart Robot Chassis Kit



Findings from Survey

- Most systems successfully detect and extinguish small fire
- Ultrasonic sensors improve autonomous navigation
- IoT-based systems provide remote monitoring advantages
- Battery efficiency and response time remain major challenges

Limitations of the Project

- Limited ability to handle large or complex fires
 - Sensor accuracy may decrease due to smoke or environmental conditions
 - Battery backup is limited for long operation
 - Obstacle avoidance may fail in crowded environments
 - Water capacity of the robot is small
- 



CONTRIBUTIONS

Team Member Name	Contribution
Abdullah AL Masud	1. Assisted in surveys to gather project requirements 2. Contributed to the construction and hardware implementation of the robot.



The background features a symmetrical arrangement of geometric shapes. On the left and right sides, there are large, overlapping triangles and squares in shades of dark blue, medium blue, and yellow. A thin, light blue line forms a stylized, angular frame around the central text area. The overall design is modern and clean.

THANK YOU